Web-Based Analytics Platform for Coastal Portfolio Optimization of Resource Technologies (C-PORT)

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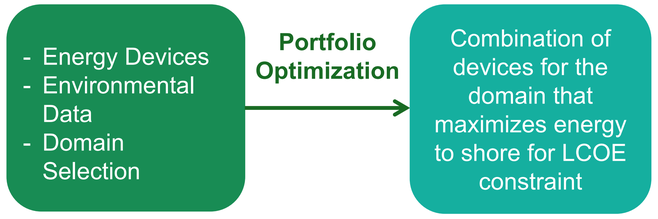
Background

- Electrification, manufacturing expansion, and AI-driven data center growth have **increased electricity demand**
- **Offshore energy can catalyze coastal economic development while meeting energy needs**



Future Market Insights Inc. (2025), International Energy Agency (IEA), Ocean Energy Systems (OES) (2023)

- Despite this opportunity, offshore wind, wave, tidal and ocean current **energy-harvesting systems** are typically **evaluated as isolated resources**
  - This fragmented approach limits planners' ability to understand how **diversified portfolios can stabilize output, reduce cost volatility, and maximize economic spillovers to coastal supply chains**
- **Coastal Portfolio Optimization of Resource Technologies (C-PORT)** framework advances energy planning from single-technology assessment to **integrated, multi-technology design**



- Uses multi-year geospatial resource datasets, technology and transmission cost and performance models, **C-PORT identifies optimal combinations of offshore technologies for a selected domain**

Aim

- **Create a web-based platform for (C-PORT)**
  - Positions NC State, our partners, and the UNC System at the **forefront of blue-economy-informed offshore energy analytics**
  - Establishes NC as a **leader in integrated marine energy planning**



Software Architecture

Frontend:
- NextJS Project
- TypeScript & Tailwind CSS
- Deployment: Vercel

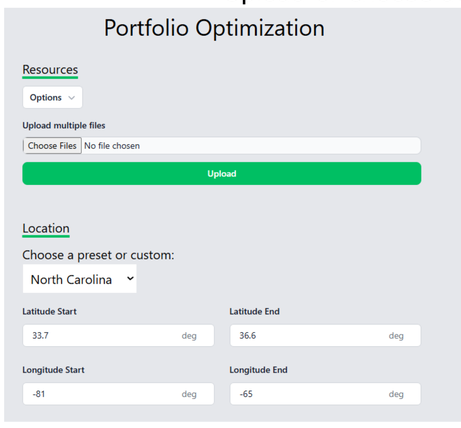
Backend:
- Flask (Python)
- API Requests: Axios

Containerization:
- Docker

Scan to see the live interface!

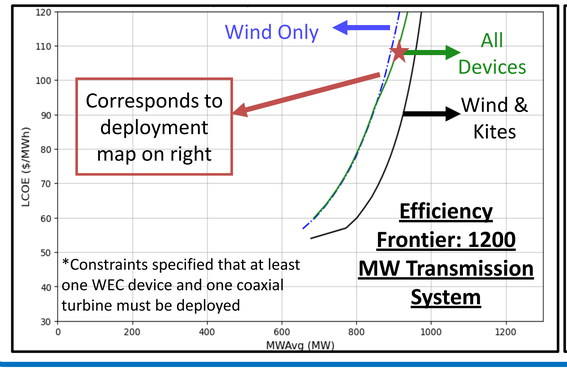
User Interface

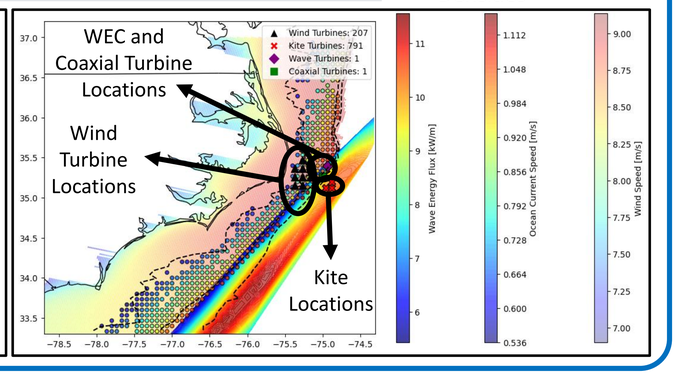
Upload or choose devices, set inputs, and generate frontiers!

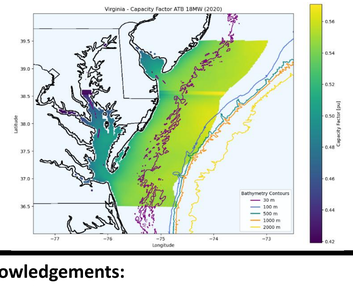


User Inputs

- Devices
- Domain
- Transmission Capacity
- Transmission System Radius
- Max Depth
- Max Distance From Shore
- LCOE Range
- LCOE Step Size
- Analysis Years
- Device Types
- Device Packing Densities







Short-Term

- Domain expansion
- Resource uploads
- Additional information output to the user
- Stakeholder investigation

WEC Cost Scenario	LCOE ($/MWh) Target	P_{avg} (MW) Wind	P_{avg} (MW) Wave	P_{avg} (MW) Kite	P_{avg} (MW) Curr.	P_{avg} (MW) Total
Baseline (100%)	100	148.3	1.95	172.44	92.72	229.98
	75	87.43	0.65	178.56	48.12	218.53
	50	18.87	0.65	187.16	12.08	194.60
75%	100	133.05	9.72	170.97	83.12	230.62
	75	87.48	0.65	178.48	48.06	218.55
	50	22.81	0.65	183.43	11.66	195.23
50%	100	102.64	36.38	161.86	67.20	233.68
	75	87.43	1.95	175.84	46.39	218.84
	50	22.81	0.65	185.83	14.53	195.72
25%	100	99.76	165.94	69.57	85.36	249.91
	75	41.82	74.58	156.44	44.07	228.77
	50	3.82	27.86	180.14	13.27	198.55

TABLE I
Table I. Optimization Results Under Different WEC Cost Scenarios

Future Work

Long-Term

- Direct commercialization pathway
- Platform deployed through subscription-based licensing, customized scenario analysis, or integration into existing energy planning workflows.

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